Part II

New Theory of Sets and Semisets

Introduction

The non-actualisability of the domain of al natural numbers, that is, the non-existence of the set of all these numbers, has far-reaching consequences for the infinitary mathematics of the 20th century. The guarantor of the existence of this set, namely the God of the medieval and baroque rational theology, who was identified as such by Bolzano and for whom no substitute has ever been found, no longer fulfills his role. And the problem is not just the non-existence of this particular set; even more importantly, most mathematicians working in theories based on Cantor's set theory and dealing with various subjects of mathematical inquiry, rely on capabilities of this divine observer, which they appropriated.

Thus the attitude of mathematicians to infinity has been as if it lay before them in its actual form; as Adam lies before God on Michelangelo's picture in the Sistine chapel.

Infinity remains even after its divine observer and maker has disappeared. We have been thrown into pre-set-theoretical mathematics. We are not above the world but within it. We are not looking for infinity in God's mind but within the world.

We are seeking that infinity which has been always present in our journeys to the boundaries of our world (or even a little beyond these boundaries). We encounter it in all directions, not just travelling to a distance but also on the way to the micro-world. In our search we can draw on mathematical knowledge and experience from the past millennia.

We shall not enter the pre-set-theoretical mathematics as it was when we left it; we shall enter it with an answer to the following question, that should be asked by every philosophically thinking mathematician.

> How is it possible that some results of classical infinitary mathematics can be applied in the natural real world[11] and others have no use even when interpreting the real world?

Contemporary infinitary mathematics is based on Cantor's theory of classical, that is, classically interpreted, (actually) infinite sets. Such sets do not exist in the natural real world and hence it is not them that infinitary mathematics applies its results to.

If some results of classical infinitary mathematics are somehow applicable in the natural real world, then this involves interpreting indistinct[12,13] phenomena

[11] By the *natural real world* we basically mean the environment whose phenomena we perceive. By the *real world*, we mean an extension of the natural real world presupposed by science.

[12] *Indistinctness* is a phenomenon of the natural real world and it should be built also into the real world, if we want to avoid Descartes' dualism. If we interpret the real world as sharply defined, then we cannot avoid declaring indistinctness to be a subjective phenomenon that has no place in the objective real world. Consequently we must grant a soul to man, which – albeit not necessarily immortal – is separated from the real world. On the other hand, if man along with his perception is a part of the real world then the phenomenon of indistinctness is is also a part of it. Moreover, indistinctness is the primary phenomenon; it is no poorly captured sharpness. On the contrary, in most cases sharpness is idealized indistinctness.

[13] "Indistinct" and "indistinctness" translates the Czech words "neostrý" and "neostrost"

of this world. For example, many results about the classical continuum are applicable when interpreting the shape of a table standing in front of us. The shape of the table is also a continuum, but a different continuum than that studied in topology by classical infinitary mathematics. Similarly, results from classical infinitary probability theory or statistics can be applied to study the indistinctly defined multitude of birds on the territory of our country at the end of summer, and so on.

If we say that these applications of classical infinitary mathematics in the natural real world are only approximate (in other words that these phenomena of the natural real world are only more or less accurately modeled in classical infinitary mathematics), we have not answered the question posed above. We have only reformulated it (or more precisely, we have reformulated its first part) as follows: How is it possible that some phenomena of the natural real world, namely the indistinct phenomena, can be modeled in classical infinitary mathematics, and sometimes even very accurately?

If infinity is applicable when interpreting indistinct phenomena of the natural real world, it must already be present in the indistinctness of these phenomena in some form. If it were not there, we could not use it in these interpretations or, if you like, these phenomena could not be modeled in classical infinitary mathematics.

The presence of infinity in indistinctness can be illustrated by the following narrative. In classical mathematics, using the non-fading effect of the principle of mathematical induction we can prove that, if we add one extra element to a finite set A, we get a set B which cannot be bijectively mapped onto the set A.[14] This proposition does not hold for infinite sets and Richard Dedekind even tried to characterize infinite sets by its invalidity. An author of a popular book demonstrated the invalidity of this proposition for sets whose elements can be numbered by all the natural numbers by the following story.

Imagine a hotel with infinitely many rooms numbered by all the natural numbers, with all the rooms occupied, each by one guest only. Let A be the set of all the hotel guests. It is still possible to accommodate another guest who has arrived to request a room, without placing two guests in one room. We can do that by putting up this guest in the room number 1 and for each n, moving the guest from the room number n to the room $n + 1$. It is obvious that we map the set B which has been created by adding one extra guest to the set A, bijectively onto the set A. Let us now imagine that the hotel has only one thousand rooms, all occupied. We can nevertheless proceed in the same way os outlined above. The newcomer is accommodated in the room number 1, the guest from the room number 1 moves to the room number 2, etc. As the whole procedure of moving guests is realised gradually, it will surely not be finished by morning (undoubtedly we will not have time to move the guest out of the room number 1000). And every guest will still have been accommodated for

respectively, which is literally “non-sharp” and “non-sharpness”. [Ed]

[14] That is, we cannot pair elements of A with elements of B so that each element of A is paired with a unique element from B and each element from B is paired with a unique element from A.

the best part of the night. In this case the set of one thousand rooms has an indistinctly defined part, or semiset, which contains all the rooms where guests change. This semiset behaves in a similar way as Cantor's classical set of all natural numbers.

To be consistent, we have to admit that infinity shows up in some form or other in the indistinctness of phenomena of the natural real world. For example, on semiset parts of large – from the classical point of view, finite – sets, not somewhere behind these large sets. We shall call this form of infinity **natural infinity**.

Of course, we could object that in the latter of the above stories, the moving of guests could be accelerated, at least theoretically. In that case we would increase the number of rooms in the hotel to match it. We would not remove the semiset. We can only make it sharper, more distinct.

The infinity we are now talking about is a sharpening of the natural infinity,[15] just as (ideal) geometric objects are sharpened shapes of objects of the natural real world.

Bolzano's programme for infinitary mathematics[16] can be expressed using the following imperative:

> Wherever infinity occurs in some form, look for a multitude on whose structure this form of infinity shows!

Bolzano's programme plays an essential role in mathematics based on Cantor's set theory. By contrast, in mathematics based on the new theory of sets and semisets the role of Bolzano's programme is only auxiliary, albeit very important. The programme of the new infinitary mathematics can be characterized by the following imperative:

> Wherever there is indistinctness, look for a horizon and natural infinity that has caused this indistinctness; then idealise this situation!

Natural real world is thus a source of immediate inspiration for the new infinitary mathematics and also the place for almost immediate applications of its results.

[15] By going back to natural infinity, we are emphasizing a crucial even if suppressed cross-reference between infinity and indistinctness.

[16] See Section 1.3. [Ed]

Chapter 5

Basic Notions

5.1 Classes, Sets and Semisets

In this introductory chapter we remind the readers of familiar notions related to classes, and we summarise some concepts important for the the new infinitary mathematics.

If we single out some previously formed objects, we will obtain a **collection** of these singled-out objects.

A **domain** is not a totality of some existing objects; it is a source or a container into which the suitable emerging or created objects fall.

Every collection of objects can be interpreted as a domain, albeit an exhausted one.

By **actualisation** of a domain we mean its exhaustion, that is, substitution of this domain by the collection of all objects that fall or can fall within it.

By a **class** we mean any collection of given objects (its elements), such that we interpret this collection as being an autonomous entity, or a single object.

By a **set** we mean a class that is sharply defined.[1] Every set is finite from the classical point of view.

By a **semiset** we mean an indistinctly defined class which is a part (that is, a subclass) of a set.

Symbols or groups of symbols used to denote objects are called **terms**.

Letters $A, B, C, \dots$ and $a, b, c, \dots$ will be used as particularly simple terms, namely as unspecified constants for objects. This means that if we say within an argument that A is an object, then A denotes the same object, no matter which one, to the very end of this argument. Consequently, the argument carried out with A applies to every object, since any object could have been denoted by this letter.

If we say that A is a constant then we mean the symbol A; if we say that A is an object then we mean the object denoted by the symbol A. One object may be denoted by various terms.

[1] Sharpness refers to distinctness, clarity and definiteness, …, in brief, the ancient perfection of geometric objects studied in Euclid's *Elements*.

The fact that two terms, for example the constants A and B, denote the same object is recorded as $A = B$ and we say that "A equals B". Even though the notation $A = B$ is sometimes read as "object A equals object B" or "object A is identical to object B", strictly speaking we always mean that symbols A and B denote the same object. The fact that two terms, for example constants A and B, denote different objects is recorded as $A \neq B$ and read "A is different from B" or "A does not equal B" etc.

The notation $A \in B$ means that the object B is a class and the object A is an element of (or belongs to) the class B.

The notation $A \notin B$ means that the object B is a class and the object A is not an element of (or does not belong to) the class B.

We say that a class A is a subclass (or a part) of a class B (notation $A \subseteq B$), if every element of the class A is also an element of the class B.

If A and B are classes for which it holds that both $A \subseteq B$ and $B \subseteq A$ then $A = B$; we create only one class from the given collection of objects (the so-called **extensionality of classes**).

An **empty class** is a class which has no elements. Extensionality implies that there is only one such class. Obviously, an empty class is a set. We will continue using the symbol $\emptyset$ for it.

The notation and definitions for classes that are unordered or ordered pairs, triples etc. of objects, or are intersections, unions etc. of other classes are as in Section 2.1. Similarly, Cartesian products of classes, definitions and properties of relations and functions along with orderings and their properties are as they were described for sets in sections 2.1.1 and 2.1.2.

5.2 Horizon

Every look[2] we cast, no matter in what direction, is limited. Either there is a firm boundary which disrupts (or deflects) it sharply, or it is limited by a **horizon** in whose direction clarity decreases and sharpness blunts.

We interpret horizon as a border separating the illuminated part of an observed object from the unilluminated part, that is, from the part not perceived by our look. In other words, by a horizon we understand only the border itself, not the illuminated part of the observed object nor all that we can capture by our look as is sometimes the case.

A look at some comprehensive object of interest is therefore connected to three distinct phenomena: the **illuminated** and the **unilluminated parts** of the object (in other words its parts lying before or beyond the horizon) and the **horizon** separating these two parts. We shall call this triple the **fundamental triad** since it will be the basis, the starting point and the reference point for all our considerations.

[2] *A look* here does not refer merely to a look by sight (with physical eyes); we understand it in a broad sense of regarding something that has been encountered.

The illuminated and unilluminated parts of the observed object cover it entirely and do not intersect; the unilluminated part of an object is defined as that from the object which does not lie within the illuminated part. The horizon belongs to none of these two parts and unlike them it is not a part of the observed object (after all, we left there no place for it to occupy).

The horizon itself, as such, is a sharp, definite and constant phenomenon.

The form of the horizon depends on the nature of a particular look. An example of a horizon that is evident[3] merely to our physical sight is the sky. In this case, the horizon shows up in an almost tangible form; this led to the sky having sometimes been interpreted as a firm and sharp boundary of the world (as a crystal surface encircling the world), and not as a horizon.

Although the horizon itself is a sharp and definite phenomenon, it does not sharply separate the illuminated part of an object from its unilluminated part. (Otherwise it would be a sharp and fixed boundary sharply and clearly delimiting the illuminated part.) The illuminated (and therefore also the unilluminated) part of an observed object is therefore delimited by the horizon indistinctly. The closer to the horizon an observed phenomenon lies, the vaguer it is. In other words, the closer we get to the horizon, the more indefiniteness there is on the illuminated part of the object. In this sense, the horizon is the place where we meet the phenomenon of indefiniteness and indistinctness in one of its purest forms.

The sharpness and definiteness of the horizon thus enable us to grasp the phenomenon of indefiniteness more definitely.

Principle of continuous transition. The illuminated part (of an object) passes continuously into the unilluminated part.

Our knowledge of this continuity does not come from our experience but, on the contrary, it precedes experience. This continuity is the very thing which distinguishes a horizon from a fixed boundary. The horizon is not some line drawn on an observed object, it is not a part of it and it is not attached to it. Therefore no phenomena belonging to the observed object mark the place where the horizon lies. Either the horizon does not touch these phenomena at all, or it separates them, too, into an illuminated and an unilluminated part. In other words, there is nothing on the observed object that would disrupt its passing from the illuminated to the unilluminated part. Naturally this does not mean that somewhere in the unilluminated part of the object (not immediately behind the horizon) there could not be a totally unexpected or even unpredictable phenomenon.

The observed object thus smoothly continues beyond the horizon until there is another phenomenon in its way which disrupts this continuity. As a consequence, the part of the observed object known to us – this **known land** (**terra cognita**) – goes beyond the illuminated part of the object, everywhere where

[3] To see something as *evident*, to evidence it, means to see it – in a wide sense of the word, not necessarily by our physical sight – and to be aware that we are seeing it.

this part is being bounded by the horizon. Naturally, this consideration does not apply to sharp and fixed boundaries of the illuminated part. What lies beyond the known land is **terra incognita**.

By the **overflow of the known land** (on an observed object, possibly also in a given direction) we mean the part by which the known land extends the illuminated part of the object; in other words, what lies simultaneously in the known land and in the unilluminated part.

The overflow of the known land therefore contains no phenomenon we would not be somehow familiar with from the illuminated part. Our far from matter-of-course confidence that our knowledge of the observed object reaches into the unilluminated part is based on this fact.

The closer to the horizon particular phenomena from the illuminated part lie, the vaguer they are and the more indistinct our knowledge of these phenomena is. Moreover, our active attention that responds to illuminated phenomena falters in the direction of the horizon. In the overflow of the known land vagueness clouds phenomena to such an extent that it is only our knowledge of their existence that remains and our awareness of these phenomena slips into passively not expecting anything unexpected.

Principle of Backward Projection. Phenomena lying on the horizon challenge us to interpret them as traces left by phenomena that have fallen beyond (or under) the horizon. More precisely: to interpret them as backward projections of these phenomena onto the places on the horizon through which there leads a way to them.

An exemplary case of this are again stars in the sky.

Phenomena lying on the horizon interpreted in this way often have no resemblance to the phenomena whose backward projections they are. They cannot be deduced from them by pure reason, not even by our wildest imagination. For example, colour is a backward projection of something that has no colour; sound is a backward projection of some waves that themselves have no sound, as witnessed by deaf people. It is the horizon that makes it possible to interpret something as a backward projection of something else; more precisely, it is the position of the horizon in the tangle of phenomena which we have been thrown into.

Therefore it is necessary to pay at least as much attention to phenomena that arise (lie) on the horizon as we pay to phenomena lying before or beyond the horizon, or to those divided by the horizon.

Scholion

Let us imagine that we could see, for example, a table standing in front of us more and more sharply. For instance, that we could observe it using an increasingly strong magnifying glass and then an increasingly strong microscope, and thus move the horizon further and further backwards. In that case we would probably soon begin to see bulges and depressions that would eventually turn into holes running through the table. The shape of the table would keep changing and eventually we would feel reluctant to call what we see a table. And it would be the same if our sight remained equally powerful but we grew smaller and smaller. In that case the observed table would gradually grow beyond the field of our vision and, should we be suddenly placed in this situation, we would not be able to recognise that we were inside some table. The observed table would disappear together with its observed shape. The preceding shapes of the table thus disappear with the horizons of our preceding looks. In other words, the shape of the observed table – and other things of course – is a phenomenon showing itself only on the horizon of a particular look.

A table without a shape is no table; it is a table only because it has that shape which shows itself to our ordinary looks into the tangle of phenomena around us. And it is the same in the case of other phenomena of the sensorily perceptible world. To put it briefly, the horizon forms our natural real world. Were it not for the horizon, our looks would bounce off rigorous boundaries or we would stare into emptiness. Our natural real world would be a cage open to nothingness in some directions; we would not inhabit a world, but a coffin. The horizon is thus the most important element of the fundamental triad; it is not an auxiliary phenomenon, but one of the fundamental footholds in the tangle of phenomena that we are thrown into.

The modern natural scientist regards phenomena on the horizon merely as a kind of a gateway into the objective real world; these phenomena are merely subjective (or inter-subjective, if they are perceived by several people). The scientist believes that objective phenomena may only be obtained if he or she manages to go beyond the horizon on which the subjective phenomena lie.

Modern natural science, which aims to explore the objective real world, thus regards phenomena on the horizon as inferior. Their role is merely auxiliary and sometimes even misleading.

However, the preceding considerations must naturally call into question whether there are any grounds for speaking about some objective shapes of various things, that is, about shapes which are independent of any looks. This would in fact only be possible if they were shapes showing on the horizon of a sharpest look (that is, a look cast by God). However, even if the aforementioned table had such an objective shape, it would not be the shape that we see now. Neither would it be a shape that we see through a microscope etc.

Moreover, there is no reason to believe that if our sight were sharper or if we grew so much smaller that we would get insight into some great depths of the microcosm, we would continue to see shapes as we are able to see them now. Perhaps we would find ourselves in a four-dimensional world and if we submerged deeper, in a ten-dimensional world, and even deeper again in a two-dimensional world twisted as a Möbius strip etc. The assumption of objective shapes is nothing other than an assumption that these changing shapes converge to some limit, or at least that different ways of sharpening our looks and submerging into the depths of the microcosm lead to limits that are in some way mutually coherent etc. However, we have no cogent argument supporting this hypothesis.

These were only a few random examples from the inexhaustible multitude of sometimes even fantastic shapes which could conceivably appear on the horizon in case that our eyesight or sense of touch continued to sharpen or if we kept growing considerably smaller; in other words, if we could cast some unnatural looks into the sensorily perceptible world. But we cannot cast such looks into this world. Even when looking through a microscope, we use our natural (that is, innate) sight, whose horizon may be brought closer or pushed back to a larger distance; but this shifting is relatively strictly limited. We do not move closer to what we observe through the microscope, we move it closer to us; we place it on the horizon of our natural look into the sensorily perceptible world, and as a consequence we see familiar shapes. We deduce from this that even in the depths of the microcosm there are the same shapes as on the horizon of our natural looks so we explain the convergence of changes in the shapes as their sharpening. There is a considerable danger of error here since everything might be completely different in the depths of the microcosm. On the other hand, we are not able to see it differently than here, and in this sense it is the same there as it is here.

From this point of view any scientific cognition of the real world is merely an organic supplementation and re-creation of phenomena appearing on a variously positioned horizon.

The famous narration about a cave that opens the seventh book of Plato's dialogue *The Republic* is usually interpreted (probably correctly) as an allegory of Plato's teaching regarding the relationship between the world of ideas and the sensorily perceptible world. Our above considerations however offer an almost realistic interpretation of Plato's enigmatic narration.

So let us recall an extract from Plato (translated by Benjamin Jowett) which will serve as the basis for the interpretation promised above.[4]

> And now, I said, let me show in a figure how far our nature is enlightened or unenlightened: – Behold! Human beings living in an underground den, which has a mouth open towards the light and reaching all along the den; here they have been from their childhood, and have their legs and necks chained so that they cannot move, and can only see before them, being prevented by the chains from turning round their heads. Above and behind them a fire is blazing at a distance, and between the fire and the prisoners there is a raised way; and you will see, if you look, a low wall built along the way, like the screen which marionette players have in front of them, over which they show the puppets.
>
> I see.
>
> And do you see, I said, men passing along the wall carrying all sorts of vessels, and statues and figures of animals made of wood and stone

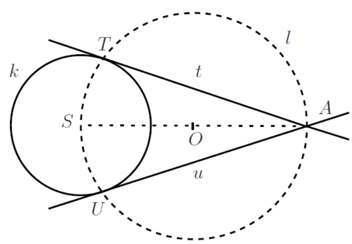
[4] Plato, *The Republic*, trans. B. Jowett, 7.514a1–515c2 (Project Gutenberg, August 2008).

> and various materials, which appear over the wall? Some of them are talking, others silent.
>
> You have shown me a strange image and they are strange prisoners.
>
> Like ourselves, I replied; and they see only their own shadows, or the shadows of one another which the fire throws on the opposite wall of the cave?
>
> True, he said; how could they see anything but the shadows if they were never allowed to move their heads?
>
> And of the objects which are being carried in like manner they would only see the shadows?
>
> Yes, he said.
>
> And, if they were able to converse with one another, would they not suppose that they were naming what was actually before them?
>
> Very true.
>
> And suppose further that the prison had an echo which came from the other side, would they not be sure to fancy when one of the passers-by spoke that the voice which they heard came from the passing shadow?
>
> No question, he replied.
>
> To them, I said, the truth would be literally nothing but the shadows of the images.
>
> That is certain.

To put it shortly, scientific cognition does not depart from the opposite wall of Plato's cave.

5.3 Geometric Horizon

The picture below shows a well-known construction of the tangents of a given circle k (with centre S) passing through a given point A lying outside the circle.



At the point when we say this, when we become aware of it, our original perception of the picture changes into geometric perception. The small dot marked by the letter T becomes an irreducible geometric point, the straight lines t, u become revealed segments of ideally straight and irreducible geometric lines, the line k – and even the broken line l – become ideal geometric circles. The line t touches the circle k at a unique point T (the point T is their only common point) and this point is simultaneously one of the two intersection points (that is, common points) of circles k, l etc. To put it briefly, our understanding of the picture now is that we are looking through it into the geometric world.

However, we will never reach the geometric world using our bodily senses. No matter how hard we try, we will never find an ideal circle through haptic reception and neither will we see it with our very own eyes. We still do look into this world, namely using our **geometric sight**, which is a kind of a spiritual sight. Using it, we can see as far as individual points and ideally straight and irreducible lines, and looking at a square we grow perfectly sure that both its diagonals are equally long. We could not pursue geometry if we were not capable of this. We could never agree on the fact that a line touches a circle at a unique point. Our imagination would fail to guide us through the geometric world when uncovering its less obvious phenomena and laws. We do not choose axioms of geometry arbitrarily but so that they are true in the geometric world which we can see by our geometric sight.

We call this world, discovered in Greek antiquity, the **intuitive** or also the **ancient geometric world.**

The lines t and k in the picture above do have some indeterminate blurred section in common. If we make them thinner and thinner, if we straighten t and smoothly round k, that is, if we abrade them smoothly, the section they have in common will grow smaller and smaller. Eventually – that is, after seeing through to the geometric world, once the line t becomes a true geometric line and the line k becomes a true circle – their common section will turn into an irreducible geometric point.

This interpretation makes it possible to place ideal geometric objects (and as a consequence the whole ancient geometric world) on the very horizon lying beyond (or below) all horizons limiting the human ability of cognition through bodily senses. To see geometrically is a special ability to see as far as this horizon by means of some spiritual sight. We call this horizon **geometric horizon**.

The horizon on which we have placed the geometric world is of a peculiar nature since it lacks some properties characteristic for a horizon. It is rigid and definite. This has enabled the pure and in the ancient Greek sense ideal science – geometry – to be set on it. Rigidity here refers to the fact that, broadly speaking, this horizon is always at the same distance from us; it cannot be pushed further away. In this respect it quite resembles the sky which is always above us, at the same distance from us, no matter whether we are standing or moving. However, in contrast to the sky, the geometric horizon is definite, namely because we can safely perceive it by means of our geometric sense. The geometric horizon could consequently become a stage for the characteristic phenomena of the geometric world situated on it. (Let us note here that as long as the sky had been interpreted in the same manner, the science of celestial phenomena had been perceived to be of equal worth to geometry in the ancient Greek sense. In other words, also this science used to have more of a claim to be knowledge of permanent and unchangeable truths than natural sciences.)

Summarizing, ancient geometry is not pursued by people, but by Olympian gods. A human can practice it only when he or she replaces his or her human outlook by that of Zeus and his or her human abilities by the abilities of a superhuman. This substitution of sensorium humanum with sensorium deorum has been made possible by the miracle of the ancient discovery of the intuitive geometric world. Geometric horizon is the horizon limiting looks cast by the Olympian gods.

The ancient geometric world and geometric perception is a gift of the Olympian gods to European humankind. Or rather, geometry is knowledge stolen from the Olympian gods. It is necessary to note here that the Europeans were punished for this theft by being bound to the intuitive geometric world with manacles even stronger than those that bound Prometheus to the rock in the Caucasus.

5.4 Finite Natural Numbers

Since the domain of all natural numbers is not actualisable, natural numbers along with their progress to the horizon, beyond the horizon and possibly as far as the depths of terra incognita, have to be approached – so to speak – from below. It means that we shall break into this domain starting from individual natural numbers and we shall actualise only those parts of it where it appears to serve some purpose. Naturally we do not intend to give up advantages obtained by actualisation of at least some parts of the domain of all natural numbers.

By natural numbers we understand numbers $0,1,2,3,\ldots$. Obviously, those which we have written down form a negligible part of the domain of all natural numbers, and this would be the case even if we wrote down the first hundred, million or trillion natural numbers. It is so even in the case of numbers smaller than 10^{421} which, according to the ancient Indian manuscript Lalitavistara, Buddha could name.

The remaining infinitely many natural numbers, that is, those which even in the wildest imagination sensorium humanum cannot comprehend, hide in those three dots (or in the words "and so on") which follow the writen-down or uttered numbers whenever we wish to acquaint somebody with natural numbers.

The remarkable thing is that these three dots (or the words "and so on") represent sufficiently, and in a way very clearly, our idea of natural numbers and invoke understanding for their progress to infinity. Since we are unable to represent natural numbers in any better way, we are undoubtedly dealing here with an archetypal idea that belongs amongst the mysterious outgrowths from the collective unconscious of – at least – Indo-European mankind (in the sense of C. G. Jung).

Similar archetypes of the collective unconscious formed those European gods which, thanks to their immortality and the unlimited penetration of their sight, are capable of reaching successively natural numbers beyond sensorium humanum.

By **finitely large** (briefly just finite) **natural numbers** we understand those natural numbers that Zeus can reach. In other words, those natural numbers from the sequence $0, 1, 2, 3, \ldots$ that lie before the geometric horizon.

If α is a finite natural number then any smaller natural number is clearly also finite. All these smaller natural numbers form a set which we denote $[\alpha]$.

Since even the geometric horizon is a horizon, the sequence of natural numbers also enters the overflow of its known land. That allows us to consider the following postulate to be at least theoretically performable.

Postulate of weak actualisability of the domain of natural numbers. For every natural number α belonging to the known land adjacent to the geometric horizon it is possible to create the set $[\alpha]$ of all natural numbers smaller than α.

The existence of at least one infinite natural number ϑ along with the fact that every finite natural number is smaller than ϑ implies the existence of the collection FN of all finite natural numbers; for the collection FN is a part of the set $[\vartheta]$. In other words, the actualisability of the domain of all finite natural numbers is a consequence of the principle of continuous transition and the postulate of weak actualisability of the domain of natural numbers.

The collection FN cannot be interpreted as a set. If FN were a set, as a subset of some $[\vartheta]$, where ϑ is an infinite natural number, it would have to have a maximum element. However, this is not possible as it is straightforward to check that if n is finite, $n + 1$ is also a finite natural number.

There is no reason why we should not view the collection FN as an independent individual, that is, to interpret this collection as an object. On the contrary, such an interpretation is undoubtedly useful.

Consequently we must conclude that the collection FN is not sharply defined. Hence the class FN is a semiset.

This means that we can use predicate calculus when studying elements of the semiset FN, their properties and relations. However, we must be very careful with the quantifiers because the semiset FN is not sharply defined in the direction of the horizon limiting the size of finite natural numbers. By contrast, predicate calculus can be used without hesitation when studying the set $[\vartheta]$, where $\mathrm{FN} \subseteq [\vartheta]$.

Chapter 6

Extension of Finite Natural Numbers

6.1 Natural Numbers within the Known Land of the Geometric Horizon

Let ϑ denote some fixed infinite natural number lying in the known land of the geometric horizon. Let $\alpha, \beta, \gamma, \ldots$ denote elements of the set $[\vartheta]$ and $m, n, k \ldots$ elements of the semiset FN. Obviously $\mathrm{FN} \subseteq [\vartheta]$.

The transition from finite natural numbers to infinite numbers is so continuous that it is impossible to determine the place where it actually happens. Thanks to the continuity of this transition, various properties of natural numbers and also many propositions and their proofs pass almost automatically from the semiset FN into the set $[\vartheta]$. For example:

(a) Natural numbers belonging to the set $[\vartheta]$ are linearly ordered according to their magnitude.

(b) Every non-empty subset of the set $[\vartheta]$ has a least and a greatest element.

(c) Arithmetical laws of natural numbers hold in the entire known land of the geometric horizon and so on.

We say that sets u, v have the same number of elements if there is a bijective function f such that f is a set and

$$\mathrm{dom}(f) = u, \quad \mathrm{rng}(f) = v.$$

Proposition 6.1. If $\alpha \in [\vartheta]$, $u \subseteq [\alpha]$ and $u \neq [\alpha]$ then the sets u and $[\alpha]$ do not have the same number of elements.

Proof.[1] Assume that $\alpha + 1 \in [\vartheta]$ is the smallest natural number for which there is $u \subseteq [\alpha + 1]$, $u \neq [\alpha + 1]$ and a bijective function f such that f is a set and

$$\mathrm{dom}(f) = [\alpha + 1], \qquad \mathrm{rng}(f) = u.$$

[1] In this proof by contradiction, we can focus on non-zero numbers since the proposition clearly holds for $\alpha = 0$. [Ed]

First let $\alpha \notin u$. Then $f(\alpha) \in u \subseteq [\alpha]$, $u \setminus \{f(\alpha)\} \neq [\alpha]$ a $f|[\alpha]$ is a bijective function mapping the set $[\alpha]$ onto the set $u \setminus \{f(\alpha)\}$, which is a contradiction.

So let $\alpha \in u$. Then $[\alpha] \cap u \neq [\alpha]$. If $f(\alpha) = \alpha$ then $f|[\alpha]$ is a bijective function mapping the set $[\alpha]$ onto the set $[\alpha] \cap u$, which is a contradiction. If $\alpha = f(\beta)$, where $\beta \in [\alpha]$, then $f(\alpha) \in [\alpha] \cap u$. We define a function g on the set $[\alpha]$ by setting $g(\beta) = f(\alpha)$ and $g(\gamma) = f(\gamma)$ for $\beta \neq \gamma \in [\alpha]$. Obviously g is a bijective function, $\mathrm{dom}(g) = [\alpha]$, $\mathrm{rng}(g) = u \cap [\alpha] \neq [\alpha]$, which is a contradiction. □

Proposition 6.2. For every set u there is at most one $\alpha \in [\vartheta]$ such that the sets u and $[\alpha]$ have the same number of elements.

Proof. Let sets $[\alpha], [\beta]$, where $\alpha, \beta \in [\vartheta]$, $\alpha \neq \beta$ have the same number of elements as a set u. Let for example $\beta \in [\alpha]$. Then also the sets $[\beta], [\alpha]$ have the same number of elements, which contradicts Proposition 6.1. □

If $\alpha \in [\vartheta]$ and sets $u, [\alpha]$ have the same number of elements then the number α is called the **number of elements of the set** u and denoted $\alpha = \mathrm{Card}(u)$.

Proposition 6.3. Let $\alpha \in [\vartheta]$, $u \subseteq [\alpha]$, $u \neq [\alpha]$. Then there exists $\beta \in [\alpha]$ such that $\beta = \mathrm{Card}(u)$.

Proof. Assume that $\alpha + 1 \in [\vartheta]$ is the smallest natural number for which there is a set $u \subseteq [\alpha + 1]$, $u \neq [\alpha + 1]$ such that $\beta = \mathrm{Card}(u)$ for no $\beta \in [\alpha]$. Let δ be the greatest element of set u (obviously $u \neq \emptyset$). Then $u \setminus \{\delta\} \subseteq [\alpha]$ and so there is some $\gamma \in [\alpha]$ such that $\gamma = \mathrm{Card}(u \setminus \{\delta\})$. Hence $\gamma + 1 = \mathrm{Card}(u)$, which is a contradiction. □

We say that a **set** u is **finite**, if there exists $m \in \mathrm{FN}$ such that $m = \mathrm{Card}(u)$. We leave it to the reader to prove the following proposition.

Proposition 6.4. (i) Let sets u, v be finite. Then also $u \cup v$ and $u \times v$, are finite sets.

(ii) Let u be a finite set whose all elements are finite sets. Then also $\bigcup u$ is a finite set.

(iii) Let u be a finite set and let $v \subseteq u$. Then also the set v is finite.

(iv) Let a relation r be a finite set. Then also the sets $\mathrm{dom}(r)$ and $\mathrm{rng}(r)$ are finite.

If we observe some set which has, for example, a million elements then various **semiset parts** of it begin to appear; for no human being has the capability to clearly see this set along with each of its element, as evident. Zeus however does have this capability. In the ancient geometric world there are no semisets, only phenomena sharpened to the limit. Since we are claiming Zeus' capabilities, we have the following:

Proposition 6.5.

(i) If α is a finite natural number and X a class such that $X \subset [\alpha]$ then X is a set.

(ii) If however $\alpha \notin \mathrm{FN}$ then FN is a semiset part of the set $[\alpha]$.

This allows us to formulate the following equivalent definition of the class of all finite natural numbers.

FN is the class of all natural numbers α such that for each $X \subset [\alpha]$, X is a set.

Scholion

The above definition of the semiset FN of all finite natural numbers can be used also when applying our theory in the real world. In such case the class FN will be much shorter than it is in the ancient geometric world. After all, line segments in the ancient geometric world are much thinner than those drawn on a piece of paper, too.

6.2 Axiom of Prolongation

Hard sets are sets in whose interior structure there is no semiset. We define them as follows.

Hard sets of type 1 are:

(a) sets of abstract ur-objects;[2]

(b) sets of natural numbers;

(c) sets of other kinds of numbers that will be discussed later.

Hard sets of type n, $n \neq 1$, are sets whose elements are only sets of smaller types than n, or abstract ur-objects, natural numbers and numbers introduced later. (For the considerations that we will be carrying out we will make do with hard sets of very small types.)

We say that a **function** F is **stable on the semiset** FN (in the following text just **stable**) if $\mathrm{dom}(F) = \mathrm{FN}$ and for every $n \in \mathrm{FN}$, the restriction of F to the set $[n]$ (that is, the function $F|[n]$) is a hard set.

Not only the transition from finite natural numbers to infinite natural numbers is so smooth that it is impossible to pinpoint the place where it actually happens. It is the same in case of various stable functions on the semiset FN. This phenomenon can be captured as follows:

Axiom of Prolongation. Let F be a stable function (on the semiset FN). Then there exists a function f such that f is a set and for every $n \in \mathrm{FN}$,

$$f(n) = F(n) \qquad \text{(that is, } F = f|\mathrm{FN}).$$

[2] In other words, objects whose contents have been emptied; they are mere objects and nothing more.

Note. Without loss of generality we can assume about f that $\mathrm{dom}\, f = [\gamma]$ where γ is an infinite natural number, $\gamma \in [\vartheta]$. If this is not the case, we set γ as the greatest element of the set of all $\beta \in dom(f)$, for which $\mathrm{dom}(f|[\beta]) = [\beta]$. Obviously $\gamma \notin \mathrm{FN}$.

It is often more convenient to use the Axiom of Prolongation in the following form:

Axiom of Prolongation. Let $\{a_n\}_{n\in\mathrm{FN}}$ be a stable sequence. Then this sequence has a (set) extension $\{a_\alpha\}_{\alpha<\gamma}$, where γ is an infinite natural number, $\gamma \in [\vartheta]$.

In mathematics based on the new (formerly alternative) theory of sets and semisets we use the Axiom of Prolongation to replace all possible forms of the notion of *limit*.

6.3 Some Consequences of the Axiom of Prolongation

Proposition 6.6. Let F be a stable function such that $F(n)$ is a set for every n (or also $F(n) \subseteq w$ respectively, where w is some given set). Then there exists an extension f of the function F such that for every $\alpha \in \mathrm{dom}(f)$, $f(\alpha)$ is a set (or also $f(\alpha) \subseteq w$ respectively) .

Proof. Let g be an extension of a function F. Let γ be the largest natural number such that $\gamma \in \mathrm{dom}(g)$ and $g(\beta)$ is a set (or also $g(\beta) \subseteq w$ respectively) for every $\beta \leq \gamma$. Evidently $\gamma \notin \mathrm{FN}$. So it suffices to set $f = g|[\gamma]$. $\square$

Proposition 6.7. Let F be a stable function such that $F(n)$ is a set for every n. Let $F(n) \subseteq F(m)$ (or $F(m) \subseteq F(n)$ respectively) for $n \leq m$.

Then there exists an extension f of the function F such that $f(\alpha)$ is a set for every $\alpha \in \mathrm{dom}(f)$ and $f(\alpha) \subseteq f(\beta)$ (or $f(\beta) \subseteq f(\alpha)$ respectively) for $\alpha \leq \beta$, $\alpha, \beta \in \mathrm{dom}(f)$.

Proof. Let g be an extension of the function F such that $g(\alpha)$ is a set for every $\alpha \in \mathrm{dom}(g)$ (see Proposition 6.6). Let γ be the largest natural number such that $\gamma \in \mathrm{dom}(g)$ and $g(\alpha) \subseteq g(\beta)$ (or $g(\beta) \subseteq g(\alpha)$ respectively) for every $\alpha < \beta < \gamma$. Evidently $\mathrm{FN} \subseteq [\gamma]$. It therefore suffices to set $f = g|[\gamma]$. $\square$

Proposition 6.8. Let F be a stable function such that $F(n)$ is a set for every n. Assume that $\bigcup \mathrm{rng}(F)$ is a set; let us denote it w. Then there exists $m \in \mathrm{FN}$ such that $w = \bigcup \mathrm{rng}(F|[m])$.

Proof. Let f be an extension of the function F such that $f(\alpha)$ is a set, $f(\alpha) \subseteq w$ for every $\alpha \in \mathrm{dom}(f)$. If $\gamma \in \mathrm{dom}(f) \setminus \mathrm{FN}$, then $w = \bigcup \mathrm{rng}(f|[\gamma])$. The smallest natural number γ such that $w = \bigcup \mathrm{rng}(f|[\gamma])$ is therefore an element of the semiset FN. $\square$

Proposition 6.9. Let F, G be stable functions such that for every m and n, $F(m)$ and $G(n)$ are sets satisfying $F(m) \cap G(n) = \emptyset$. Then there exist sets u, v such that $u \cap v = \emptyset$ and for every n

$$F(n) \subseteq u, \quad G(n) \subseteq v.$$

Proof. Let f, g be extensions of the functions F, G such that $f(\alpha)$, $g(\alpha)$ are sets for every $\alpha \in \mathrm{dom}(f) \cap \mathrm{dom}(g)$ (see Proposition 6.6). Let γ be the largest natural number such that $\gamma \in \mathrm{dom}(f) \cap \mathrm{dom}(g)$ and $f(\alpha) \cap g(\beta) = \emptyset$ for every $\alpha, \beta \in [\gamma]$. Evidently $\gamma \notin \mathrm{FN}$. If we set

$$u = \bigcup \mathrm{rng}(f|[\gamma]), \qquad v = \bigcup \mathrm{rng}(g|[\gamma]),$$

then u, v are sets with the desired properties. □

Proposition 6.10. Let F, G be stable functions such that $F(n)$, $G(n)$ are sets satisfying $F(n) \cap G(n) \neq \emptyset$ for every n and $F(m) \subseteq F(n)$, $G(m) \subseteq G(n)$ for every $n \leq m$. Then there exists x such that $x \in F(n) \cap G(n)$ for every n.

Proof. Let f, g be extensions of functions F, G. Let γ be the largest natural number such that $[\gamma + 1] \subseteq \mathrm{dom}(f) \cap \mathrm{dom}(g)$ and $f(\alpha)$, $g(\alpha)$ are sets satisfying $f(\alpha) \cap g(\alpha) \neq \emptyset$ for every $\alpha \leq \gamma$ and for every $\alpha < \beta \leq \gamma$ we have $f(\beta) \subseteq f(\alpha)$, $g(\beta) \subseteq g(\alpha)$. Evidently $\gamma \notin \mathrm{FN}$. If $x \in f(\gamma) \cap g(\gamma)$, then $x \in f(n) \cap g(n) = F(n) \cap G(n)$ for every n. □

Proposition 6.11. Let F be a stable function such that $F(n)$ is a non-empty set for every n. Let $F(m) \subseteq F(n)$ for every $n \leq m$. Then there exists x such that $x \in F(n)$ for every n.

Proof. This proposition is a special case of Proposition 6.10, namely for $F = G$. □

If $\gamma \in [\vartheta]$ is an infinite number then evidently also for every n, $\gamma - n$ is an infinite number.

Proposition 6.12. Let $\{\gamma_n\}$ be a sequence of infinite numbers smaller than ϑ such that $\gamma_{n+1} \leq \gamma_n$ for every n. Then there exists an infinite number β such that $\beta < \gamma_n$ for every n.

Proof. Evidently $[n] \cap ([\vartheta] \setminus [\gamma_n]) = \emptyset$ for every n. According to Proposition 6.9 there exists a set v such that $\mathrm{FN} \subseteq v$ and $v \cap ([\vartheta] \setminus [\gamma_n]) = \emptyset$ for every n. Let us choose $\beta \in v \setminus \mathrm{FN}$. Then $\beta \notin [\vartheta] \setminus [\gamma_n]$ for every n and so $\beta \in [\gamma_n]$, in other words $\beta < \gamma_n$. □

6.4 Revealed Classes

It this section, we consider only classes whose every element is either an abstract ur-object or a hard set. A class of such classes will be called a **class cluster**.

We remark that the Axiom of Prolongation may not apply to a class cluster (sequence) Z. For example the class cluster $\{\mathrm{FN} \setminus [n]\}_{n\in\mathrm{FN}}$ has no suitable extension.

Similarly to previous sections, ϑ denotes a fixed infinite number lying in the known land of the geometric horizon.

Whenever a set is considered in this section, it means a set X such that $\mathrm{Card}(X) \in [\vartheta]$.

Let F be a stable function, $\mathrm{rng}(F) \subseteq X$. We say that f is a **set extension of the function** F in the class X if

(i) f is a set and a function,

(ii) $F \subseteq f$,

(iii) $\mathrm{rng}(f) \subseteq X$,

(iv) $\mathrm{dom}(f) = [\gamma]$ for some $\gamma \notin \mathrm{FN}$.

We say that a **class** X is **revealed** if every stable function F such that $\mathrm{rng}(F) \subseteq X$ has a set extension in the class X.

In other words, a class X is revealed if every sequence $\{a_n\}_{n\in\mathrm{FN}}$ such that $a_n \in X$ for every n has a set extension $\{a_\alpha\}_{\alpha\leq\gamma}$ where γ is an infinite natural number and $a_\alpha \in X$ for every $\alpha \leq \gamma$.

In very broad terms, a revealed class is such a class from which there is no way out (not even for Zeus).

The following proposition is an immediate consequence of Proposition 6.6.

Proposition 6.13. (i) A class X is revealed if and only if for every sequence $\{a_n\}_{n\in\mathrm{FN}}$ such that $a_n \in X$ for every n there exists a set $v \subseteq X$ such that $a_n \in v$ for every n.

(ii) Every set X is revealed.

Proposition 6.14. Let classes X, Y be revealed. Then also the class $X \times Y$ is revealed.

Proof. Let $\{\langle x_n, y_n\rangle\}_{n\in\mathrm{FN}}$ be a sequence of elements of the class $X \times Y$. Then obviously there are sets u, v such that $u \subseteq X$, $v \subseteq Y$ and $x_n \in u$, $y_n \in v$ for every n. It means that $\langle x_n, y_n\rangle \in u \times v \subseteq X \times Y$ for every n. $\square$

Proposition 6.15. Let classes X, Y be revealed. Then also the class $X \cup Y$ is revealed.

Proof. Let $\{a_n\}_{n\in\mathrm{FN}}$ be a sequence of elements of the class $X \cup Y$. If the entire sequence lies in the class X (or the entire sequence lies in the class Y) then there is nothing to prove. Let $a_k \in X$, $a_l \in Y$ and define

$$a'_n = \begin{cases} a_n & \text{for } a_n \in X, \\ a_k & \text{for } a_n \notin X, \end{cases} \qquad a''_n = \begin{cases} a_n & \text{for } a_n \in Y, \\ a_l & \text{for } a_n \notin Y. \end{cases}$$

Let u, v be sets such that $u \subseteq X$, $v \subseteq Y$ and $a'_n \in u$, $a''_n \in v$ for every n. Then obviously $a_n \in u \cup v \subseteq X \cup Y$ for every n. □

Proposition 6.16. Let classes X, Y be revealed. Then also the class $X \cap Y$ is revealed.

Proof. Let $\{a_n\}_{n \in \mathrm{FN}}$ be a sequence of elements of the class $X \cap Y$. Let $\{a_\alpha\}_{\alpha \leq \gamma_1}$ where $\gamma_1 \notin \mathrm{FN}$ (or $\{a_\alpha\}_{\alpha \leq \gamma_2}$ where $\gamma_2 \notin \mathrm{FN}$ respectively) be its set extensions in the class X (or Y respectively). Let β be the greatest natural number for which the two extensions agree up to β. Then obviously $\beta \notin \mathrm{FN}$ and $\{a_\alpha\}_{\alpha \leq \beta}$ is a set extension of the sequence $\{a_n\}_{n \in \mathrm{FN}}$ in the class $X \cap Y$. □

Silent Version of the Axiom of Choice. Let F be a function such that $\mathrm{dom}(F) = \mathrm{FN}$ and $F(n)$ is a non-empty class for every n. Then there exists a function G such that $\mathrm{dom}(G) = \mathrm{FN}$ and $G(n) \in F(n)$ for every n.

This is a very weak version of the Axiom of Choice implicitly used by mathematicians since long before the birth of set theory without it being seen as problematic. In fact, it was accepted by some opponents of the general Axiom of Choice even later on; for example, in the proof that two different definitions of continuity of a real function at a given point are equivalent. That is why we too will be using this axiom without mentioning it; in fact we might have implicitly used it already.

By induction for finite natural numbers we can easily prove that the above postulated **choice function** G is stable. This enables us to state the silent Axiom of Choice as follows.

Proposition 6.17. Let F be a function, $\mathrm{dom}(F) = \mathrm{FN}$ and let $F(n)$ be a non-empty class for every n.

Then there exists a function f such that f is a set, $\mathrm{dom}(f) = [\gamma]$ where $\gamma \notin \mathrm{FN}$, $\gamma \in [\vartheta]$ and $f(n) \in F(n)$ for every n.

Proposition 6.18. Let G be a function such that $\mathrm{dom}(G) = \mathrm{FN}$ and $G(n)$ is a revealed class for every n. Then the class

$$X = \bigcap \{G(n); n \in \mathrm{FN}\}$$

(the intersection of all the classes $G(n)$) is revealed.

Proof. Let $\{a_n\}_{n \in \mathrm{FN}}$ be a sequence of elements of the class X. Let $\{a_\alpha\}_{\alpha \leq \gamma}$ where $\gamma \notin \mathrm{FN}$ be a set extension of it. Let $H(n)$ denote the class of all $\alpha \leq \gamma$ such that $a_\alpha \in G(n)$. If $\{\overline{a}_\alpha\}_{\alpha \leq \delta}$ is a set extension of the sequence $\{a_n\}_{n \in \mathrm{FN}}$ in the class $G(n)$ then the smallest β such that $\overline{a}_\beta \neq a_\beta$ satisfies $\beta \notin \mathrm{FN}$. Hence $H(n) \setminus \mathrm{FN} \neq \emptyset$.

By the Silent Version of the Axiom of Choice there exists a sequence $\{\beta_n\}_{n \in \mathrm{FN}}$ of natural numbers such that $\beta_n \in H(n) \setminus \mathrm{FN}$ for every n. By Proposition 6.12 there exists β such that $\beta \notin \mathrm{FN}$ and $\beta \leq \beta_n$ for every n. In consequence, the sequence $\{a_\alpha\}_{\alpha \leq \beta}$ is a set extension of the sequence $\{a_n\}_{n \in \mathrm{FN}}$ in every class $G(n)$, and so also in the class X. □

Proposition 6.19. Let G be a function such that $\mathrm{dom}(G) = \mathrm{FN}$ and $G(n)$ is a non-empty revealed class for every n. Let $G(n) \subseteq G(m)$ for $m \leq n$. Then the class

$$X = \bigcap\{G(n); n \in \mathrm{FN}\}$$

is non-empty.

Proof. By the Silent Version of the Axiom of Choice there exists a sequence $\{a_n\}_{n\in\mathrm{FN}}$ such that $a_n \in G(n)$ for every n. If $n \leq m$, $a_m \in G(n)$. Let $\{a_\alpha\}_{\alpha\leq\gamma}$ be a set extension of this sequence. The class $G(n)$ is revealed so there exists $\beta_n \notin \mathrm{FN}$ such that $a_\alpha \in G(n)$ for $n \leq \alpha \leq \beta_n$. Again let $\beta \notin \mathrm{FN}$ be such that $\beta \leq \beta_n$ for every n. Then $a_\beta \in G(n)$ for every n, and so $a_\beta \in X$. □

Proposition 6.20. Let relation R be a revealed class. Then also the classes $\mathrm{dom}(R)$, $\mathrm{rng}(R)$ are revealed.

Proof. Let $\{a_n\}_{n\in\mathrm{FN}}$ be a sequence of elements of the class $\mathrm{dom}(R)$. By the Silent Version of the Axiom of Choice there exists a sequence $\{b_n\}_{n\in\mathrm{FN}}$ such that $\langle b_n, a_n\rangle \in R$ for every n. Let $\{\langle b_\alpha, a_\alpha\rangle\}_{\alpha\leq\gamma}$ be a set extension of the sequence $\{\langle b_n, a_n\rangle\}_{n\in\mathrm{FN}}$ in the class R. Then obviously $\{a_\alpha\}_{\alpha\leq\gamma}$ is a set extension of the sequence $\{a_n\}_{n\in\mathrm{FN}}$ in the class $\mathrm{dom}(R)$.

The case of $\mathrm{rng}(R)$ is analogical. □

Proposition 6.21. Let G be a function such that $\mathrm{dom}(G) = \mathrm{FN}$. Let $G(n)$ be a revealed class for every n. Assume that $G(0)$ is a relation and $G(m) \subseteq G(n)$ for $n \leq m$. Let $X = \bigcap\{G(n); n \in \mathrm{FN}\}$. Then

$$\mathrm{dom}(X) = \bigcap\{\mathrm{dom}\, G(n); n \in \mathrm{FN}\}.$$

Proof. Obviously $\mathrm{dom}(X) \subseteq \mathrm{dom}\, G(n)$ for every n. Conversely let x be such that $x \in \mathrm{dom}\, G(n)$ for every n. Let $S(n) = \{y; \langle y, x\rangle \in G(n)\}$. $S(n)$ is a non-empty revealed class for every n and $S(n) \subseteq S(m)$ whenever $m \leq n$. Hence there is some y such that for every n, $y \in S(n)$ so $\langle y, x\rangle \in G(n)$ and consequently $\langle y, x\rangle \in X$. Hence $x \in \mathrm{dom}(X)$. □

6.5 Forming Countable Classes

We say that a **class** X is **countable** if there exists a bijective function F such that

$$\mathrm{dom}(F) = \mathrm{FN} \qquad \text{and} \qquad \mathrm{rng}(F) = X.$$

We say that a **class** X is **finite** if $n \in \mathrm{FN}$ and there exists a bijective function F such that

$$\mathrm{dom}(F) = [n] \qquad \text{and} \qquad \mathrm{rng}(F) = X.$$

If X is a finite class then X is a finite set since the collection of its elements is sharply defined by the function F.

We say that a **class** X is **uncountable** if it is not finite nor countable.

We say that a **class** X is **at most countable** if it is either countable or finite.

Proposition 6.22. Let X be a subclass of the class FN. Then the following holds:

(i) Let $X \neq \emptyset$. Then the class X has a first element.

(ii) Let $0 \in X$ and let $n + 1 \in X$ for every $n \in X$. Then $X = \mathrm{FN}$.

Proof. (i) Choose $n \in X$. Then $[n] \cap X$ is a set. If this set is empty, then n is the first element of the class X; if this set is non-empty, then its first element is obviously also the first element of the class X.

(ii) Assume that $X \neq \mathrm{FN}$. Let n be the first element of the class $\mathrm{FN} \setminus X$ (it exists according to (i)). Since $n \neq 0$, there exists m such that $n = m + 1$. Obviously $m \in X$ and therefore also $n \in X$, which is a contradiction. □

Proposition 6.23. Let $X \subseteq \mathrm{FN}$ and assume that for ever $n \in X$ there exists $m \in X$ such that $n < m$. Then the following is true:

(i) For every $n \in X$ there exists $m \in X$ such that $\mathrm{Card}(X \cap [m]) = n$.

(ii) There exists a bijective function F for which

$$\mathrm{dom}(F) = \mathrm{FN}, \qquad \mathrm{rng}(F) = X$$

and if $n < m$ then $F(n) < F(m)$; hence the class X is countable.

Proof. (i) For a proof by contradiction, assume that n is the smallest natural number for which there exists no $m \in X$ such that $\mathrm{Card}(X \cap [m]) = n$. Obviously $n \neq 0$ and therefore $n = n_0 + 1$. Let $m_0 \in X$ be a number such that $\mathrm{Card}(X \cap [m_0]) = n_0$. Let $m \in X$ be the smallest number such that $m_0 < m$. Then $\mathrm{Card}(X \cap [m]) = n$, which is a contradiction.

(ii) For $n \in X$, let us define $G(n) = \mathrm{Card}(X \cap [n])$. Obviously G is a bijective function and $\mathrm{dom}(G) = X$. By (i), $\mathrm{rng}(G) = \mathrm{FN}$ and for $m, n \in X$, $m < n$ if and only if $G(m) < G(n)$. The function F, which is inverse to the function G, has all the desired properties. □

It is straightforward to check that the following holds.

Proposition 6.24. Let G be a bijective function,

$$X = \mathrm{dom}(G), \qquad Y = \mathrm{rng}(G).$$

If one of the classes X, Y or G is countable, the remaining two are also countable. Similarly, if one of the classes X, Y or G is finite, the remaining two are also finite.

If X is a countable class, $Y \subseteq X$, then the class Y is at most countable.

Proposition 6.25. Let F be a function such that $\mathrm{dom}(F) = \mathrm{FN}$ and $F(n)$ is a hard set for every $n \in \mathrm{FN}$. Then the function F is stable.

Proof. Let X be a class of all n for which $F|[n]$ is a set. Obviously $0 \in X$, since $F|\emptyset = \emptyset$. Let $n \in X$. Then

$$F|[n+1] = F|[n] \cup \{\langle F(n), n\rangle\}; \qquad \text{hence} \quad n+1 \in X.$$

By Proposition 6.22, $X = \mathrm{FN}$. □

We clearly have

Proposition 6.26. Let Z be a countable class. Then Z is not a set.

Proposition 6.27. (i) Let X be a countable class, Y at most countable. Then the class $X \cup Y$ is countable.

(ii) Let A be a finite cluster of countable classes. Then $\bigcup A$ is a countable class.

Proof. (i) Let F_1 and F_2 be bijective functions,

$$\begin{aligned} &\mathrm{dom}(F_1) = X, && \mathrm{rng}(F_1) = \mathrm{FN}, \\ &\mathrm{dom}(F_2) = Y, && \mathrm{rng}(F_2) \subseteq \mathrm{FN}. \end{aligned}$$

Let

$$F(x) = \begin{cases} 2F_1(x) & \text{for } x \in X, \\ 2F_2(x)+1 & \text{for } x \in Y \setminus X. \end{cases}$$

Then F is obviously a bijective function,

$$\mathrm{dom}(F) = X \cup Y, \qquad \mathrm{rng}(F) \subseteq \mathrm{FN}.$$

Since the class $\mathrm{rng}(F)$ is countable, the class $\mathrm{dom}(F)$ is also countable.

(ii) For a proof by contradiction, let n be the smallest natural number such that there is a cluster A of countable classes such that $\mathrm{Card}(A) = n$ but $\bigcup A$ is not a countable class.

Obviously $A \neq \emptyset$; let us choose $Y \in A$. Let us set $B = A \setminus \{Y\}$. Since $\mathrm{Card}(B) < n$, the class $\bigcup B$ is countable (or empty). Hence by (i), $\bigcup A = \bigcup B \cup Y$ is a countable class, which is a contradiction. □

Proposition 6.28. Let F be a function such that $\mathrm{dom}(F)$ is a countable class. Then the class $\mathrm{rng}(F)$ is at most countable.

Proof. Without loss of generality, we can assume that $\mathrm{dom}(F) = \mathrm{FN}$. Let X be a class of all n for which the following holds: if $m < n$ then $F(m) \neq F(n)$. Obviously $F|X$ is a bijective function, $\mathrm{rng}(F) = \mathrm{rng}(F|X)$. Since the class X is either countable or finite, the class $\mathrm{rng}(F)$ is also at most countable. □

Proposition 6.29. (i) The class $\mathrm{FN} \times \mathrm{FN}$ is countable.

(ii) Let X be a countable class, $Y \neq \emptyset$ at most countable. Then $X \times Y$ is a countable class.

Proof. (i) Define $G(\langle m, n\rangle) = 2^m \cdot 3^n$. It is straightforward to check that G is a bijective function, $\mathrm{dom}(G) = \mathrm{FN} \times \mathrm{FN}$, $\mathrm{rng}(G) \subseteq \mathrm{FN}$. The class $\mathrm{rng}(G)$ is countable because it is not finite, and therefore the class $\mathrm{dom}(G)$ is also countable.

(ii) Let F_1 and F_2 be bijective functions,

$$\begin{aligned} &\mathrm{dom}(F_1) = X, && \mathrm{rng}(F_1) = \mathrm{FN}, \\ &\mathrm{dom}(F_2) = Y, && \mathrm{rng}(F_2) \subseteq \mathrm{FN}. \end{aligned}$$

For $\langle x, y\rangle \in X \times Y$ let

$$F(\langle x, y\rangle) = \langle F_1(x), F_2(y)\rangle.$$

Obviously F is a bijective function, $\mathrm{dom}(F) = X \times Y$, $\mathrm{rng}(F) \subseteq \mathrm{FN} \times \mathrm{FN}$. The class $\mathrm{rng}(F)$ is countable because it is not finite. As a consequence, the class $\mathrm{dom}(F)$ is also countable. □

Various functions defined on the semiset FN (and therefore also various countable classes) can be generated by means of the so-called construction by mathematical induction. One relatively general version of this principle is contained in the following proposition.

Proposition 6.30. (On construction of a sequence by induction) Let $\mathcal{O}$ be some unary sharply defined operation that turns any finite sequence $\{z_i\}_{i<n}$ into an object $\mathcal{O}(\{z_i\}_{i<n})$. Then there exists a unique sequence $\{v_n\}_{n\in\mathrm{FN}}$ such that for each n, $v_n = \mathcal{O}(\{v_i\}_{i<n})$.

Proof. First we will prove that there exists at most one sequence with the desired property. Assume that $\{v_n\}_{n\in\mathrm{FN}}$, $\{u_n\}_{n\in\mathrm{FN}}$ are two such different sequences. Let n be the smallest natural number such that $u_n \neq v_n$. Then $v_n = \mathcal{O}(\{v_i\}_{i<n}) = \mathcal{O}(\{u_i\}_{i<n}) = u_n$, which is a contradiction.

Now we shall prove that at least one such sequence does exist. Let A be the class of all n for which there exists a finite sequence $\{u_i\}_{i<n}$ such that $u_k = \mathcal{O}(\{u_i\}_{i<k})$ for every $k < n$. Analogously to the first case, we can now prove that for every $n \in A$ there exists only one such finite sequence. If $n \in A$ then, if we add one more term $u_n = \mathcal{O}(\{u_i\}_{i<n})$ to the sequence $\{u_i\}_{i<n}$ corresponding to number n, we get a desired sequence for $n+1$ again. Hence $n+1 \in A$. Consequently $A = \mathrm{FN}$.

It suffices to set $v_n = u_n$, where $\{u_i\}_{i<n+1}$ is the sequence corresponding to the number $n+1$. Obviously $\{v_n\}_{n\in\mathrm{FN}}$ is as required. □

Although the construction by induction is a very useful tool for creating countable classes, it is far from being a sufficient tool for this purpose. We have witnessed this when circumstances forced us to accept the so-called Silent Version of the Axiom of Choice.

Those variants of the Axiom of Choice that are sufficient for the new (alternative) theory of sets and semisets and are both sufficiently weak and transparent that they would have been acceptable in mathematics even before set theory, can be summarized in the axiom of horizon attainability stated below.

As it is in the case of the Axiom of Prolongation, this axiom is also an axiom in the original sense of the word. This means an axiom as something that should be accepted, not an axiom as the word is used today.

We say that a **function** f is **short** if there exists $n \in \mathrm{FN}$ so that

$$\mathrm{dom}(f) = [n].$$

Axiom of Horizon Attainability. Let $Z \neq \emptyset$ be a class whose every element is a short function, and such that if $f \in Z$, $\mathrm{dom}(f) = [n]$ then there exists $g \in Z$ satisfying $f \subseteq g$ and $\mathrm{dom}(g) = [n+1]$. Then there exists a function F such that $\mathrm{dom}(F) = \mathrm{FN}$ and $F|[n] \in Z$ for every n.

Scholion

One of the incentives for the formulation and acceptance of the Axiom of Horizon Attainability is the following consideration.

The ability to make a decision, even if we value the options from which we choose almost equally, is one of the characteristic human features. In other words, no remotely sensible person would behave as the legendary donkey that perished on a dusty road because he could not make up his mind whether to assuage his unbearable hunger grazing on the field to the right or on the field to the left.

Naturally, a super-human cannot be denied this ability either, not even in the case where s/he must choose from possibilities that are of entirely equal worth as regards some abstract object of study. This would, for example, be the case in the following story.

Assume that a super-human wants to walk to the horizon through an environment where after each step s/he needs to select a place to set foot from many possible places. S/he selects one of them and on taking the step s/he is faced with a similar decision again. Such an environment is passable for the super-human. And usually there exists more than one direct way leading through as far as the horizon.

Proposition 6.31. Let Y be an infinite class. Then there exists a countable class X such that $X \subseteq Y$.

Proof. Let Z be the class of all bijective short functions f such that $\mathrm{rng}(f) \subseteq Y$. If $f \in Z$, $\mathrm{dom}(f) = [n]$, then the class $Y \setminus \mathrm{rng}(f) \neq \emptyset$. Let us choose $y \in Y \setminus \mathrm{rng}(f)$ and set $g = f \cup \{\langle y, n\rangle\}$. Obviously $g \in Z$. The class Z satisfies the requirements of the Axiom of Horizon Attainability. Let F be a function guaranteed to exist by this axiom. If we set $X = \mathrm{rng}(F)$ then X is as required. □

Now we have the means to prove the Silent Version of the Axiom of Choice from the Axiom of Horizon Attainability.

Proposition 6.32. Let G be a function such that $\mathrm{dom}(G) = \mathrm{FN}$ and $G(n)$ is a non-empty class for every n. Then there exists a function F such that $\mathrm{dom}(F) = \mathrm{FN}$ and $F(n) \in G(n)$ for every n.

Proof. Let Z be the class of all short functions f such that if $\mathrm{dom}(f) = [n]$, then $f(m) \in G(m)$ for each $m < n$. It is straightforward to check that the class Z meets the requirements of the Axiom of Horizon Attainability; if F is a function guaranteed to exist by this axiom then F has the required properties. □

Proposition 6.33. Let A be a countable class cluster. Assume that for every $X \in A$, X is at most countable. Then the class $\bigcup A$ is at most countable.

Proof. Obviously we can suppose that $X \neq \emptyset$ for every $X \in A$. Let $A = \{X_n\}_{n \in \mathrm{FN}}$ and let us choose $\gamma \in \mathrm{N} \setminus \mathrm{FN}$. Define $B = \{Y_n\}_{n \in \mathrm{FN}}$, where Y_n is a class of all functions f such that $\mathrm{dom}(f) = [\gamma]$ and $X_n = \mathrm{rng}(f|\mathrm{FN})$. According to the Silent Version of the Axiom of Choice there exists a function G such that $\mathrm{dom}(G) = \mathrm{FN}$ and $G(n) \in Y_n$ for every n. Let $g_n = G(n)$. Let F be a function such that

$$\mathrm{dom}(F) = \mathrm{FN} \times \mathrm{FN} \qquad \text{and} \qquad F(\langle n, m\rangle) = g_n(m).$$

Obviously $\bigcup A = \mathrm{rng}(F)$ and according to Proposition 6.28, $\bigcup A$ is a finite or countable class. □

6.6 Cuts on Natural Numbers

As before, ϑ denotes a fixed infinite natural number lying in the known land of the geometric horizon. Letters α, β, γ denote elements of the set $[\vartheta]$, letters $m, n \ldots$ elements of the semiset FN.

The letter M will be used to denote a headless cut on set $[\vartheta]$ such that $\mathrm{FN} \neq M$.[3]

Proposition 6.34. Let a set f be a function for which the following holds:

(i) $\mathrm{dom}(f) = [\gamma]$ where $\gamma \notin \mathrm{FN}$;

(ii) $\mathrm{rng}(f) \subseteq [\vartheta]$;

(iii) $f(n) \in \mathrm{FN}$ for every $n \in \mathrm{FN}$;

(iv) $\alpha < f(\alpha)$ for every $\alpha < \gamma$;

(v) if $\alpha < \beta < \gamma$, then $f(\alpha) < f(\beta)$.

Then there exists a cut M such that $M \subseteq [\gamma]$ and $f(\alpha) \in M$ for every $\alpha \in M$.

[3] This means that for every $\alpha, \beta < \vartheta$, $\alpha < \beta$, if $\beta \in M$, then also $\alpha \in M$; and $M = \{\alpha; \alpha < \gamma\}$ for no γ.

Proof. We form a sequence $\{\gamma_n\}_{n\in\mathrm{FN}}$ by induction, setting $\gamma_1 = \gamma$ and choosing γ_{n+1} to be the largest natural number such that $f(\gamma_{n+1}) < \gamma_n$. For every $n \in \mathrm{FN}$ obviously $\gamma_n \notin \mathrm{FN}$, and

$$M = \bigcap\{[\gamma_n]\}_{n\in\mathrm{FN}}$$

is a cut with the required properties. □

Proposition 6.35. Let γ be an infinite natural number $\gamma < \vartheta$. Then there exist cuts $M_1, M_2, M_3 \subseteq [\gamma]$ different from FN such that the following holds:

(i) $2\alpha \in M_1$ for every $\alpha \in M_1$;[4]

(ii) $\alpha^2 \in M_2$ for every $\alpha \in M_2$;[5]

(iii) $2^\alpha \in M_3$ for every $\alpha \in M_3$.

Proposition 6.36. Let M be a cut and assume that M is not a union of countably many sets. Then the class M is revealed.

Proof. Let $\{\alpha_n\}_{n\in\mathrm{FN}}$ be a sequence of numbers that belong to the class M. Obviously $\bigcup\{[\alpha_n]\}_{n\in\mathrm{FN}} \subseteq M$ and hence there exists $\gamma \in M$ such that $\{\alpha_n\}_{n\in\mathrm{FN}} \subseteq [\gamma]$. The sequence $\{\alpha_n\}_{n\in\mathrm{FN}}$ has an extension in the set $[\gamma]$, and therefore also in the class M. □

4 So if $\alpha, \beta \in M_1$, then also $\alpha + \beta \in M_1$.

5 So if $\alpha, \beta \in M_2$, then also $\alpha\beta \in M_2$.